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SOFTWARE REQUIREMENT SPECIFICATION (SRS) and

SOFTWARE DESIGN SPECIFICATION (SDS)

TripConnect

CSD024

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B.TECH CSE V SEMESTER

Chapter 1 SRS

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1. INTRODUCTION

TripConnect is a comprehensive web-based travel platform designed to simplify and enhance the travel planning experience. The system allows users to create customized trips by selecting destinations, travel dates, budgets, and personal preferences, offering flexibility to travellers who wish to design their own journeys. At the same time, the platform provides a wide range of inbuilt travel packages curated to suit different travel types such as solo, family, or group trips, ensuring convenience for users who prefer ready-made options.

The website also integrates matchmaking features, enabling travellers to connect with like-minded companions, along with secure payment processing for bookings and refunds. Additional functionalities such as messaging, notifications, reviews, and travel history make TripConnect an all-in-one travel companion. By combining customization and pre-designed packages, TripConnect bridges the gap between independent travel planning and professional tour services, providing a seamless, user-friendly, and efficient solution for modern travellers.

1.1 Purpose

- Simplify the travel planning process for users through a centralized platform that supports both customized trip creation and inbuilt package selection.
- Enable travellers to create profiles, set preferences, and connect with compatible travel companions for shared journeys.
- Hassle-Free Bookings – Quick access to predefined packages with clear details for easy decision-making.

1.2 Scope

***1.* Software Product to be Produced**

tripConnect – a web-based travel platform that helps users find travel companions, plan customized trips, and explore predefined packages.

***2.* What the Software Product Will Do**

- Let users create profiles with travel preferences.
- Match users with suitable travel buddies.
- Enable customized trip planning and provide predefined packages.
- Support trip booking, secure payments, and notifications.
- Allow communication between travellers and include reviews/ratings.

3. **Application and Benefits**

- Collaboration & Connectivity – enables users to connect with like-minded travellers and plan trips together.
- Personalization – provides options to design customized itineraries or choose from predefined packages.
- Centralized Access – keeps all trip data, bookings, and preferences in one place for easy access.
- Efficiency & Transparency – simplifies the travel planning process with clear cost and itinerary details.
- Cost Sharing & Safety – allows users to share expenses with companions while ensuring safer journeys.
- User-Friendly Interface – offers an intuitive and easy-to-use design, making trip planning simple for all users.

1.3 Definitions, Acronyms, and Abbreviations.

DEFINATIONS-

TripConnect - A digital platform that centralizes travel planning, bookings, and social networking for travellers.

Admin - The system administrator responsible for managing users, travel packages, and system settings within TripConnect.

User - Any individual who uses TripConnect to plan, book, or network for travel purposes.

Matchmaking Engine - A feature that connects users with other travellers who have similar preferences, schedules, or destinations.

Customized Trip - A travel itinerary designed according to a user's personal preferences, including choice of destinations, activities, duration, and budget.

Package - A predefined travel itinerary available on TripConnect, including destinations, activities, schedules, and pricing.

ACRONYMS

Acronym	*Expanded Form*
DOB	*Date of Birth*
API	*Application Programming Interface*

UI	*User Interface*
DB	*Database*
GDPR	*General Data Protection Regulation*
SSL	*Secure Socket Layer*
OTP	*One Time Password*
UX	*User Experience*

ABBREVIATION

Abbreviation	***Description***
UID	*Unique identifier for each user*
TID	*Unique identifier for each trip*
PID	*Identifier for transaction or payment*
MID	*Identifier used to track matchmaking*
DID	*Identifier for a location or place*
NOTIF	*Alert or message sent to users*
ADM	*Administrator of the platform*

1.4 Overview

The remainder of this SRS provides a detailed description of TripConnect.

Section 2 gives an overall description of the platform in simple terms, focusing on how it benefits travellers and the key functions it provides. This section is mainly for customers, stakeholders, and potential users who want to understand the system's purpose without technical depth.

Section 3 then defines the functional and non-functional requirements in detail, which are more relevant to developers and technical stakeholders. These requirements specify how the system will be built, managed, and executed to deliver the expected features.

2. The Overall Description

TripConnect is a web-based platform that helps travellers find companions, plan personalized trips, and explore predefined packages. It connects users based on preferences like destination, budget, and travel style, while supporting features such as profile management, trip creation, matchmaking, messaging, payments, reviews, and notifications.

2.1 Product Perspective

TripConnect is an independent and self-contained web-based system designed to connect travellers and provide customized as well as predefined travel options. Unlike standalone booking websites (which mainly focus on tickets, hotels, or package sales), TripConnect emphasizes community-driven travel by matching users with potential companions and enabling group travel.

The product also differentiates itself from generic travel forums and social platforms by integrating profile management, matchmaking, trip planning, messaging, reviews, payments, and notifications into a single, centralized system.

2.1.1 Product Function

TripConnect is a travel management platform designed to simplify planning, booking, and social networking for travellers. The major functions of the system are summarized below in customer-friendly language.

Major Functions

1-User Registration and Profile Management

- Allow users to create, update, and manage their personal profiles.
- Store travel preferences, past trips, and social connections.

2-Package Browsing and Booking

- Display predefined travel packages with clear details (destinations, activities, schedules, pricing).
- Enable users to book packages easily through the platform.

3-Customized Trip Planning

- Allow users to create personalized travel itineraries based on their preferences, budget, and schedule.

4-Matchmaking, Search, and Filtering

- Connect users with other travellers who have similar interests, destinations, or travel dates.
- Suggest potential travel companions to enhance shared travel experiences.
- Provide advanced search and filtering options or packages, destinations, and activities based on budget, duration, interests, or location.

5-Notifications and Alerts

- Notify users of booking confirmations, package updates, or messages from matched travellers.

6-Admin Management

- Allow administrators to manage users, packages, and system settings.

- Monitor user activity and maintain platform integrity.

7-Payment Integration

- Facilitate secure online payments for package bookings and customized trips.

2.1.2 Hardware Interfaces

TripConnect is a software platform accessed via standard computing and mobile devices. The system does not require any specialized hardware. The hardware interface considerations are as follows:

User Devices

Supported devices include desktops, laptops, tablets, and smartphones.

Interfaces are standard input/output devices: touchscreen, keyboard, mouse.

Networking Hardware

Requires standard network connectivity (Wi-Fi, 4G/5G, Ethernet) to access the platform.

Communication occurs over standard web protocols (HTTP/HTTPS).

Payment Processing

Interacts with online payment gateways via the internet.

No direct connection to physical card readers or other specialized payment hardware is needed.

Conclusion: TripConnect has no dedicated hardware interface requirements beyond standard computing devices and network connectivity.

2.1.3 Software Interfaces

MongoDB – Serves as the primary database for storing user details, packages, groups, payments, chats, and travel history. Communication is handled via the MongoDB Node.js driver with data exchanged in BSON/JSON format.

Stripe API – Provides secure online payment processing for bookings, cancellations, and refunds. Interaction occurs through REST APIs over HTTPS, with requests and responses formatted in JSON, ensuring safe and reliable transactions.

Google Maps API – Enables TripConnect to display destinations, routes, and itineraries. The system communicates with the API using HTTPS requests and receives JSON-formatted responses to visualize trip details.

2.1.4 Communications Interfaces

The tripConnect system will use standard internet protocols to communicate with users and external services. The main communication interfaces are:

- PROTOCOLS – TCP/IP, HTTP

2.2 User Characteristics

The main users of tripConnect are travellers who want to find companions, plan trips, or book travel packages. Their key characteristics are:

1. **Educational Level**

- Users may be students, working professionals, or regular travellers.
- They do not need special knowledge, just the ability to understand simple online instructions.

2. **Experience with Technology**

- Most users know how to use websites, create accounts, and make online payments.
- Some may not be very comfortable with technology, so the website should be simple and easy to use.

3. **Technical Expertise**

- Users are not expected to have any technical skills.
- The system should guide them with clear menus, easy steps, and simple error messages.

4. **Impact on Design**

- The website must have a clean, user-friendly interface so that anyone can use it.
- Important features like booking, payments, and messaging should be very straightforward.
- Notifications and reminders should use clear, simple language.

2.3 Constraints

While developing tripConnect, there are some limits that we need to follow:

1. Regulatory Policies – The system must follow privacy and data protection rules to keep user information safe.

2. Hardware Limitations – It should work smoothly on normal computers and smartphones with internet access.

3. Interface to Other Applications – The system must connect with payment gateways (for bookings and refunds) and with SMS/Email services (for notifications).

4. **Parallel Operation** – The system should support many users at the same time without slowing down.

5. Audit Functions – Important actions like logins, bookings, and payments should be recorded for safety and checking later.

6. Reliability Requirements – The system should be available most of the time so users can depend on it.

7. Criticality of the Application – Since it handles money and personal data, it must be very secure and trustworthy.

8. Safety and Security Considerations – User data must be protected using encryption and secure login methods. Unauthorized access should not be possible.

2.4 Technologies used

The TripConnect website will be built using the following key technologies:

FRONTEND TOOLS- React.js / Redux / Material UI / Axios.

BACKEND TOOLS- Node.js / Express.js / Mongoose / MongoDB.

OTHER TOOLS AND TECHNOLOGIES- Stripe API / Twilio / Socket IO / Navigation / AI-ML

3. Specific Requirements

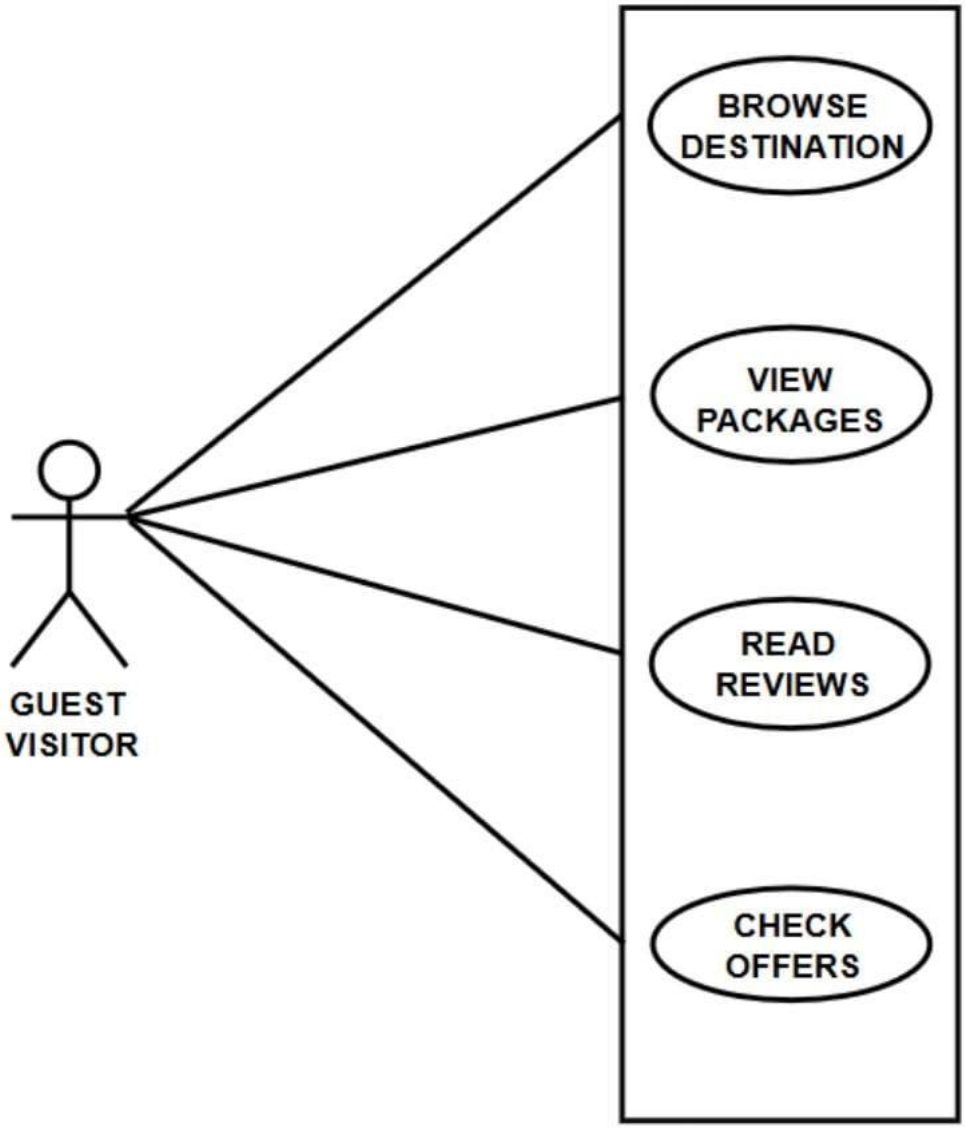
This section outlines the detailed functional and non-functional requirements for the **Trip Connect** web-based travel platform. These requirements define the core behaviour of the system, user interactions, system constraints, and performance expectations. Each requirement is presented in a way that adheres to the SRS principles of correctness, unambiguity, completeness, consistency, verifiability, modifiability, and traceability.

3.1 Functional Requirements

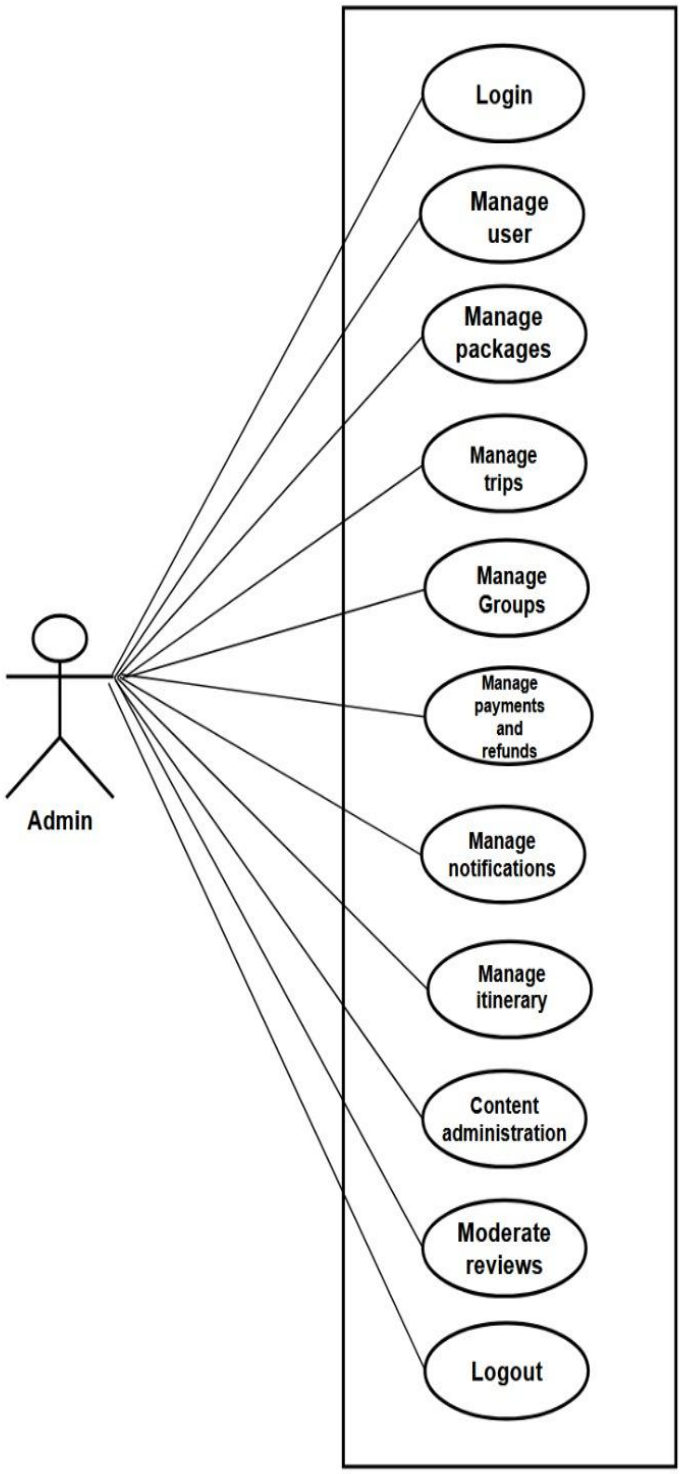
TripConnect shall allow users to register, log in, and manage profiles. The system shall enable users to customize trips or choose predefined packages, form groups through matchmaking, and communicate via chat and notifications. It shall support secure payment processing, itinerary generation, and refund handling. Admins shall manage users, trips, packages, payments, and reviews while ensuring group approval and system content updates.

3.1.1 USE CASE DIAGRAMS

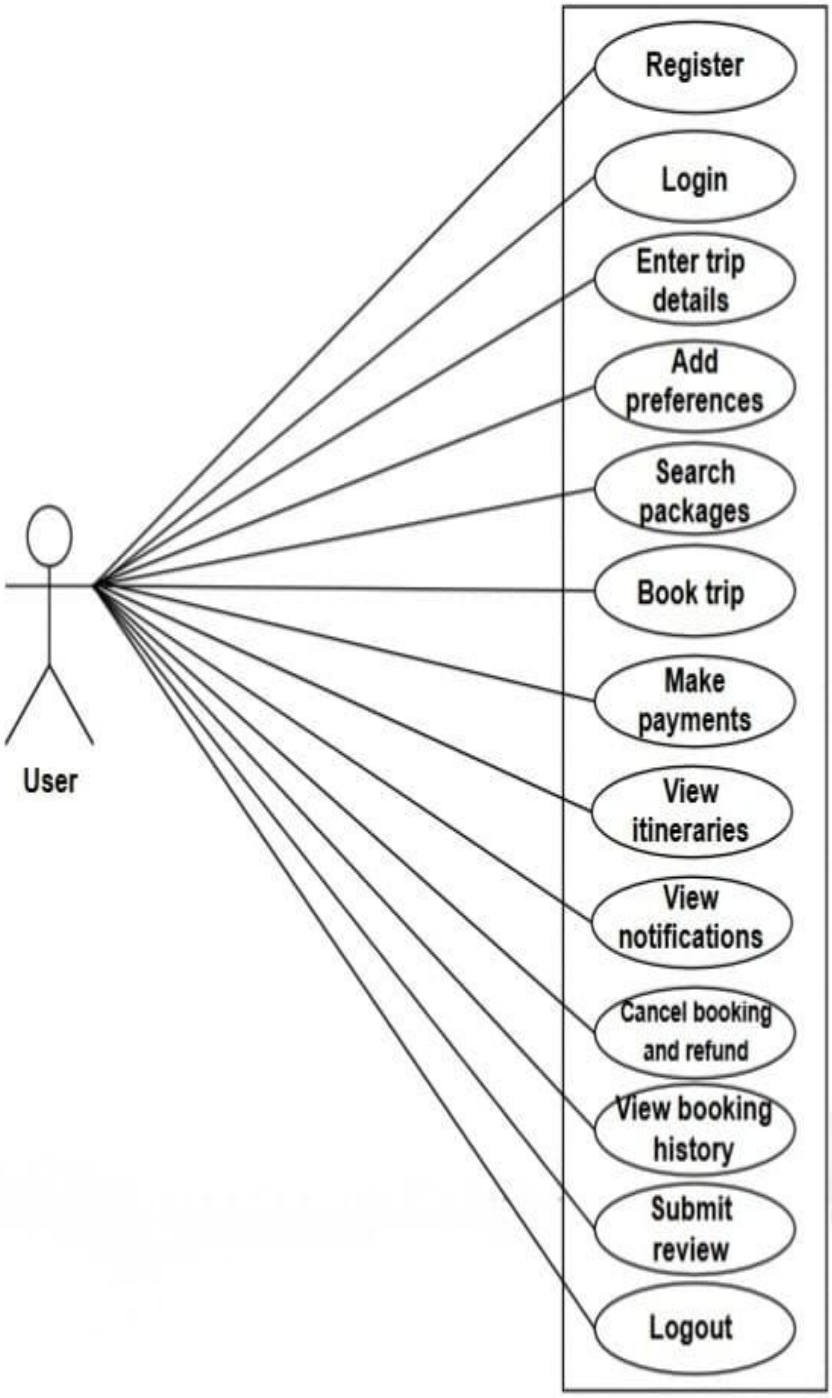
1. GUEST VISITOR



2. ADMIN


Admin
Login
Manage user
Manage packages
Manage trips
Manage Groups
Manage payments and refunds
Manage notifications
Manage itinerary
Content administration
Moderate reviews
Logout

3. USER


User
Register
Login
Enter trip details
Add preferences
Search packages
Book trip
Make payments
View itineraries
View notifications
Cancel booking and refund
View booking history
Submit review
Logout

3.1.2 USE CASE DESCRIPTION

- **ADMIN**

USE CASE NO.	1
USE CASE NAME	Login
ACTOR	Admin
DESCRIPTION	Allows Admin to log into the system
INPUT	Username, Password
NORMAL COURSE EVENTS	• Admin enters credentials • System validates • Admin logged in
ALTERNATE COURSES	Invalid credentials -> Error shown
OUTPUT	Redirected to Admin dashboard

USE CASE NO.	2
USE CASE NAME	Manage User
ACTOR	Admin
DESCRIPTION	Admin can add,update,delete,or view users
INPUT	User details (Name, Email, Phone, Role, Status)

NORMAL COURSE EVENTS	**Admin selects action → System updates database**
ALTERNATE COURSES	**User not found / Duplicate user → Error**
OUTPUT	**User successfully managed**

USE CASE NO.	**3**
USE CASE NAME	**Manage Trips**
ACTOR	**Admin**
DESCRIPTION	**Admin oversees trips created by users**
INPUT	**Trip ID, Trip details**
NORMAL COURSE EVENTS	**Admin views trips → Updates or cancels if required**
ALTERNATE COURSES	**Trip not found → Error shown**
OUTPUT	**Trip updated/cancelled**

USE CASE NO.	**4**
USE CASE NAME	**Manage Groups**
ACTOR	**Admin**
DESCRIPTION	**Admin manages travel groups**
INPUT	**Group details (Name, Members, Trip ID)**

NORMAL COURSE EVENTS	**Admin adds/updates/deletes groups**
ALTERNATE COURSES	**Group not found → Error**
OUTPUT	**Group successfully managed**

USE CASE NO.	**5**
USE CASE NAME	**Manage payments & refunds**
ACTOR	**Admin**
DESCRIPTION	**Admin monitors payments and processes refunds**
INPUT	**Payment ID, refund request**
NORMAL COURSE EVENTS	**Admin verifies payment/refund->Approves/Rejects**
ALTERNATE COURSES	**Invalid payment ID-> error shown**
OUTPUT	**Payment/refund processed successfully**

USE CASE NO.	**6**
USE CASE NAME	**Content Administrator**
ACTOR	**Admin**
DESCRIPTION	**Admin manages website/app content (FAQs, policies, banners, etc.).**

INPUT	**Content details**
NORMAL COURSE EVENTS	**Admin updates content → System publishes**
ALTERNATE COURSES	**Invalid content format → Error**
OUTPUT	**Content updated successfully**

- **USER**

USE CASE NO.	**1**
USE CASE NAME	**Register**
ACTOR	**User**
DESCRIPTION	**Allows new users to create an account**
INPUT	**Full name, Email, Phone no, Username, Password**
NORMAL COURSE EVENTS	• **User enters details** • **System Validates** • **Account created**
ALTERNATE COURSES	**Email/username already exists -> shows error**
OUTPUT	**Registration successful**

USE CASE NO.	**2**
USE CASE NAME	**Enter trip details**

ACTOR	User
DESCRIPTION	User provides travel details for planning
INPUT	Destination, dates, number of travellers, budget
NORMAL COURSE EVENTS	• User fills trip details • System stores info
ALTERNATE COURSES	Invalid date/empty fiels->error
OUTPUT	Trip details saved

USE CASE NO.	3
USE CASE NAME	Add preferences
ACTOR	User
DESCRIPTION	User adds travel preferences
INPUT	Hotel type, Activities, Food preferences
NORMAL COURSE EVENTS	• User selects preferences • System stores them
ALTERNATE COURSES	Invalid option → error
OUTPUT	Preferences saved

USE CASE NO.	4
USE CASE NAME	Book trip
ACTOR	User
DESCRIPTION	User books a trip package
INPUT	Selected package details, traveller info
NORMAL COURSE EVENTS	• User selects a package- • Confirms booking • System reserves
ALTERNATE COURSES	Package not available -> error
OUTPUT	Booking confirmation

USE CASE NO.	5
USE CASE NAME	Search packages
ACTOR	User
DESCRIPTION	User searches available travel packages
INPUT	Destination, budget, dates
NORMAL COURSE EVENTS	• User enters filters • System displays matching packages

ALTERNATE COURSES	**No packages found->show "no results"**
OUTPUT	**List of packages displayed**

USE CASE NO.	**6**
USE CASE NAME	**Logout**
ACTOR	**User**
DESCRIPTION	**Ends user session**
INPUT	**Logout request**
NORMAL COURSE EVENTS	• **User clicks logout** • **Session terminated**
ALTERNATE COURSES	**Session timeout auto logout**
OUTPUT	**User logged out**

3.2 Non-functional Requirements (Software System Attributes)

3.2.1 Availability

The system will be available online as long as the user has an active internet connection. Registered users can access all features after logging in, while non-registered users can only view limited information such as homepage content and package details.

3.2.2 Security

The system will ensure secure login and data protection using authentication and authorization. User passwords will be stored in encrypted form. Different roles (User, Admin) will have different levels of access. Unwanted or unauthorized attempts will be blocked.

3.2.3 Reliability

The system will maintain data backups so that in case of crashes or server issues, user data remains safe and can be recovered without loss.

3.2.4 Portability

The system will be developed as a web-based application using React.js and Node.js, making it accessible from any modern browser on desktops, laptops, or mobile devices without additional installation.

3.2.5 Maintainability

The website will follow a modular structure (frontend, backend, database separated), making it easier to update, fix bugs, and add new features without affecting the whole system.

3.2.6 Flexibility

The system will be flexible enough to support future enhancements like adding new features

Chapter 2 SDS

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1. Introduction

The purpose of this Software Design Specification (SDS) is to define what the tripConnect system must do to meet user and business needs. tripConnect is a web-based platform that allows travellers to connect with each other, plan customized trips, and explore predefined travel packages.

This document outlines the system's scope, functions, constraints, and interactions with users and external services. It does not describe how the system will be implemented but instead provides a clear foundation for designers and developers to build the solution effectively.

1.1 Purpose

The purpose of this Software Design Specification (SDS) is to clearly define what the tripConnect system must do in order to support its goal of connecting travellers, enabling customized trip planning, and offering predefined travel packages. This document translates the requirements from the Software Requirements Specification (SRS) into a structured form that will guide the system's design and development.

1.2 Scope

Software Product

The software product to be developed is tripConnect, a web-based platform that enables travellers to connect, plan customized trips, and explore predefined travel packages.

What the Product Will and Will Not Do

Will Do: The system will allow users to register, create and manage profiles, set travel preferences, plan trips, find compatible travel buddies, make secure payments, communicate via real-time chat, receive notifications, and provide ratings/reviews. Admins will be able to manage users, trips, and packages.

Will Not Do: The system will not act as a full travel agency or provide offline services; it is designed primarily as a connection and planning platform.

Application

The application is a web-based travel companion and trip management platform. It leverages matchmaking, personalization, and secure payment features to simplify the travel planning process.

Benefits, Objectives, and Goals

Benefits: Provides a safer and easier way for travellers to find companions, collaborate on trips, and handle payments securely.

Objectives: To centralize trip planning, communication, and package management in one platform.

Goals: To deliver a user-friendly, reliable, and secure system that enhances collaboration, transparency, and enjoyment in travel planning.

1.3 Definitions, Acronyms, and Abbreviations.

ADMIN

- Administrator is the controller of the system who manages user accounts, trip details, packages, and overall platform security.

DATABASE (DBMS)

- A structured collection of data stored electronically. In this project, it stores user details, trips, preferences, and reviews. The Database Management System (DBMS) is the software used to manage and interact with the database.

UI (User Interface)

- The visual part of the system that users interact with, including pages, buttons, forms, and menus.

ACTIVITY DIAGRAM

- A UML diagram showing the flow from one activity to another in the system, such as login, trip creation, and matchmaking.

DFD (Data Flow Diagram)

- A diagram representing the flow of data between different parts of the system, helping visualize inputs, processes, and outputs.

ER MODEL (Entity Relationship Model)

- A diagram illustrating how entities like users, trips, and payments relate to each other in the database.

SDS

- Software Design Specification – a document that describes what the system must do to guide developers in building it.

1.4 Overview

The SDS document provides a detailed blueprint for the development of tripConnect, a web-based platform that connects travellers, enables customized trip planning, and offers predefined travel packages. It is structured to guide the design and development team in building a reliable, user-friendly, and secure software solution.

The SDS Contains:

1. System Architectural Design: A visual representation of the system architecture is provided to help readers grasp the system's organization.

2. Detailed Description of Components: Detailed information is presented, including data flow diagrams, use case diagrams, entity-relationship diagrams, activity diagrams.

3. Structure and Relationships: This section clarifies the interrelationships and dependencies between modules, layers, and components

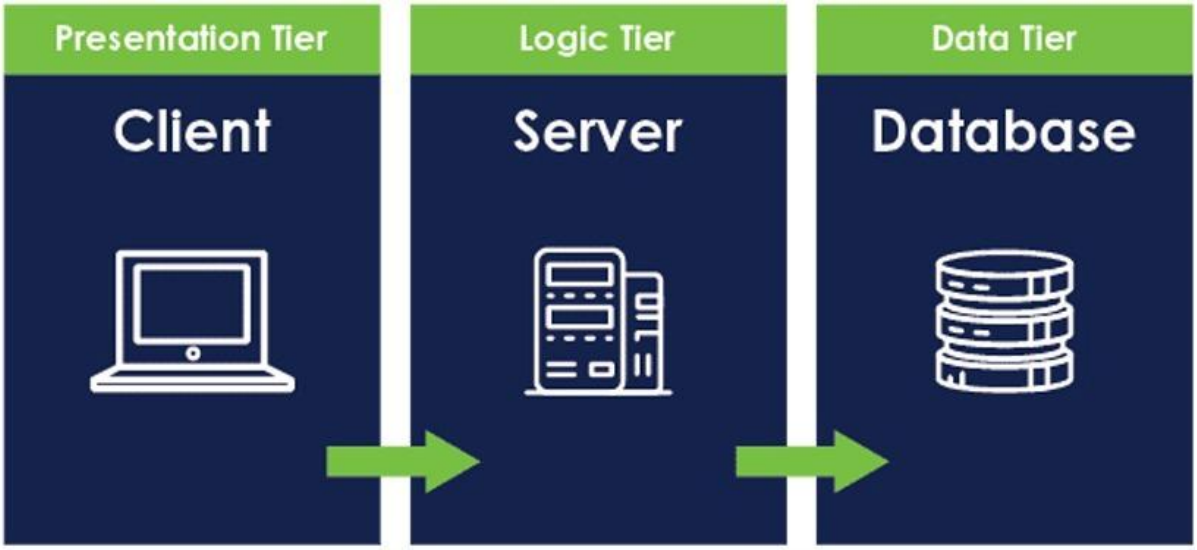
4. Data Design: This subsection provides insight into data structures, encompassing internal, global, and temporary data structures.

5. Database Description: It outlines the database tables, their purposes, methods of use, field descriptions, data types, and integrity constraints.

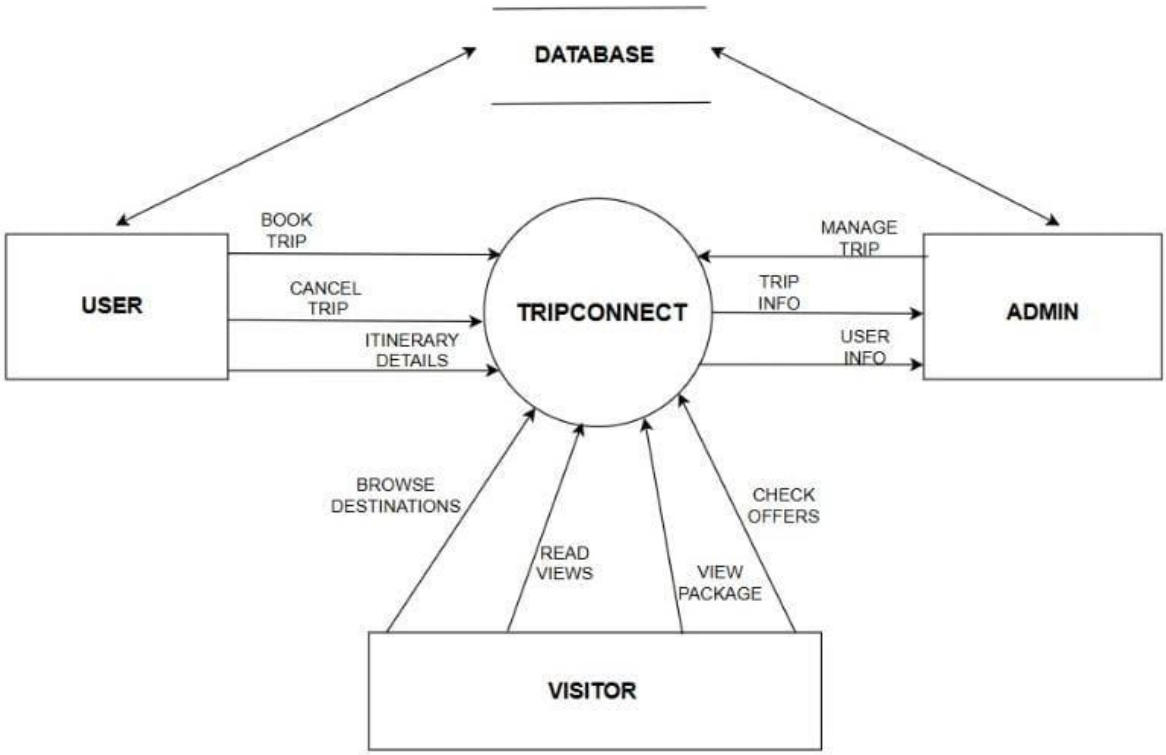
6. User Interface Design:Objects and actions within the interface are identified, describing their purpose, data types, properties, and responses to different events.

7. Types of Tests: It includes Unit Tests, System Tests, Integration or Regression Tests, Stress Tests, Acceptance Tests, and Staging.

2.1 High Level Design Overview



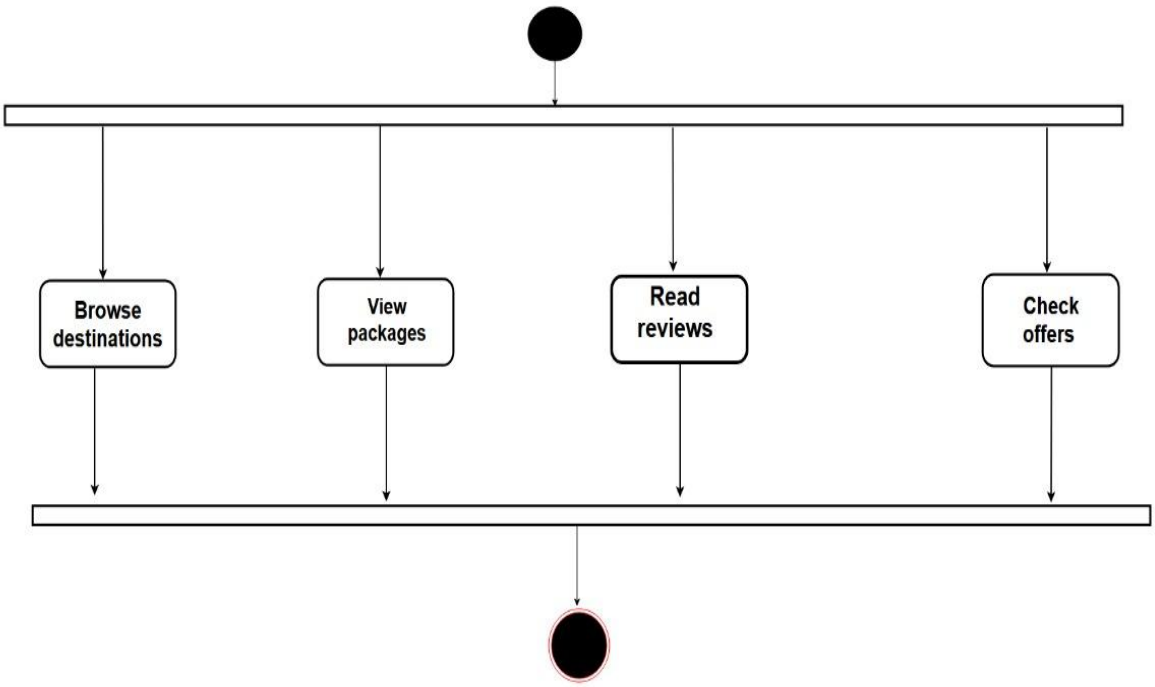
DFD 0-LEVEL



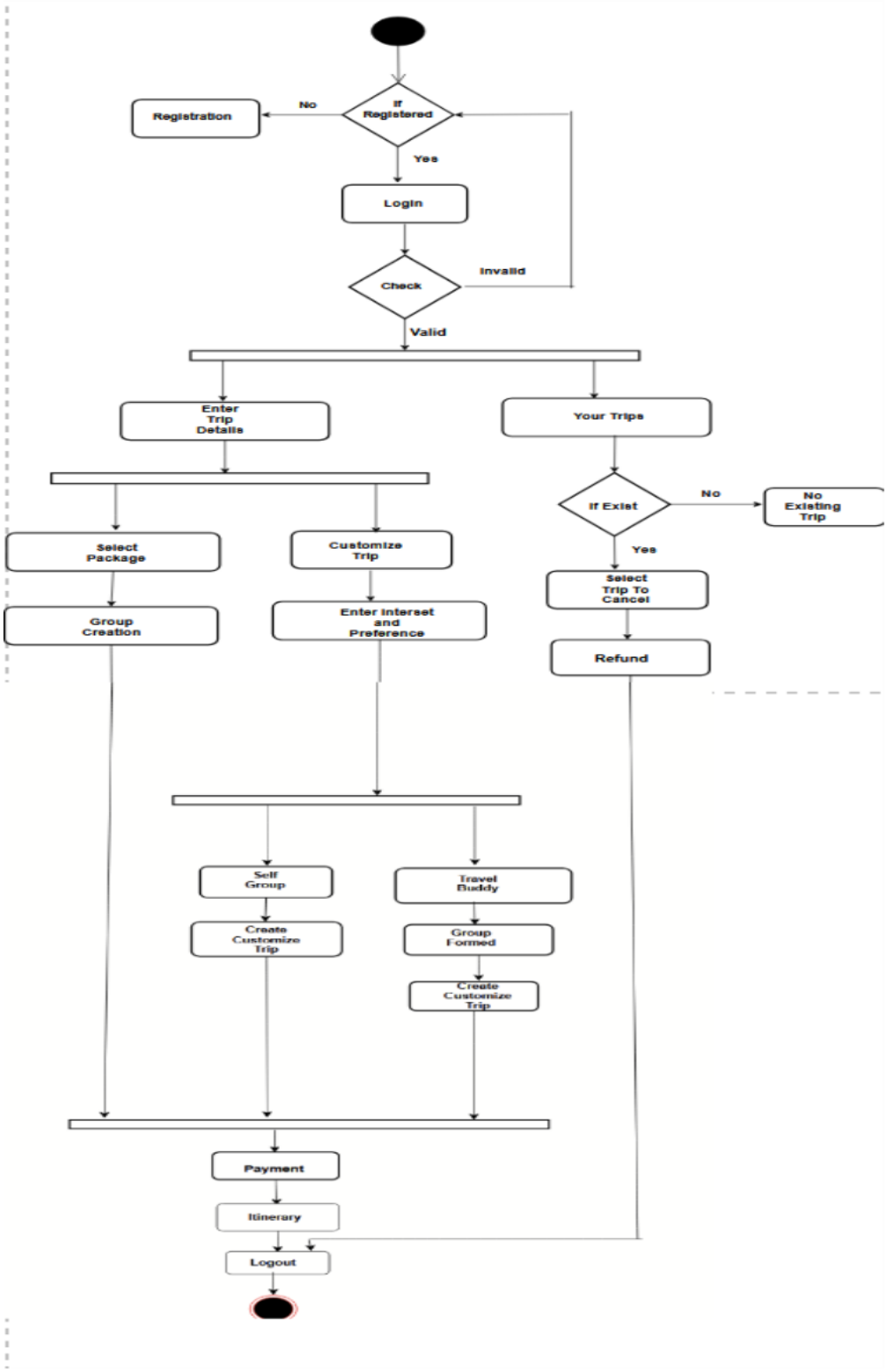
ACTIVITY DIAGRAM

An activity diagram is a UML diagram that shows the flow of activities and actions in a system or process.
It visually represents step-by-step workflows, decisions, and parallel processes.

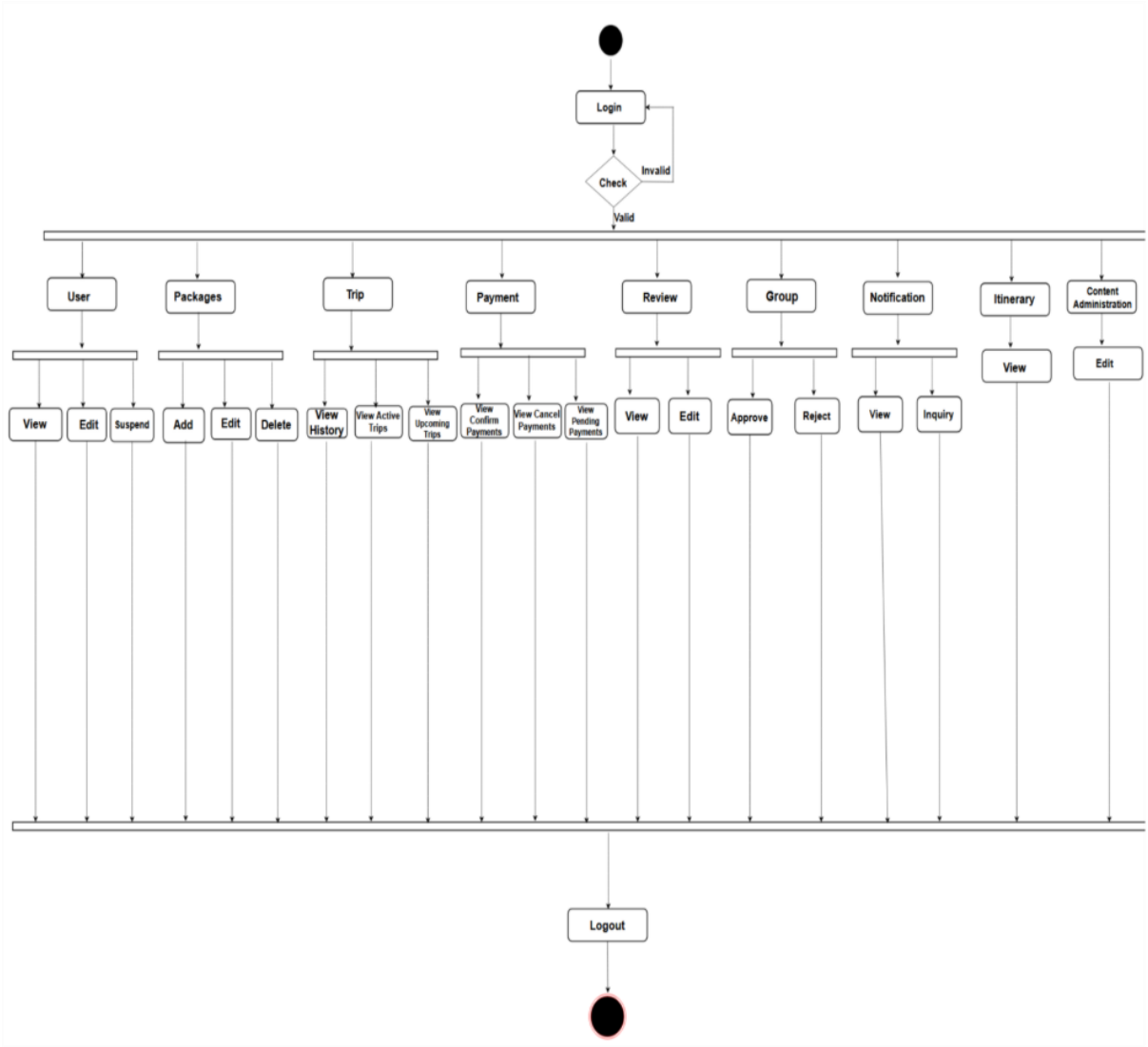
1. VISITOR



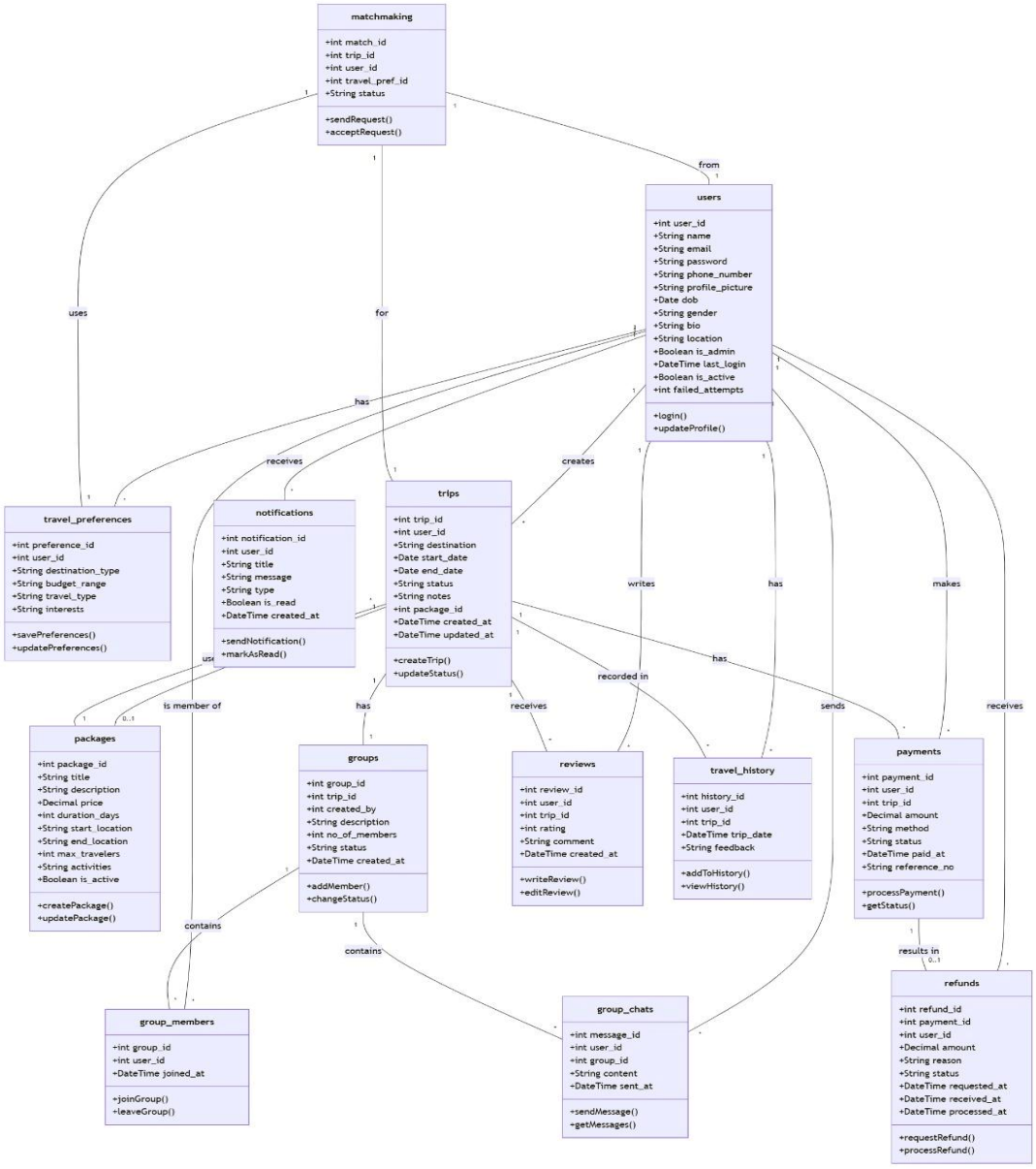
2. FOR USER


Registration
No
If Registered
Yes
Login
Check
Invalid
Valid
Enter Trip Details
Your Trips
If Exist
No
No Existing Trip
Yes
Select Trip To Cancel
Refund
Select Package
Customize Trip
Group Creation
Enter Interset and Preference
Self Group
Travel Buddy
Create Customize Trip
Group Formed
Create Customize Trip
Payment
Itinerary
Logout

- **ADMIN**

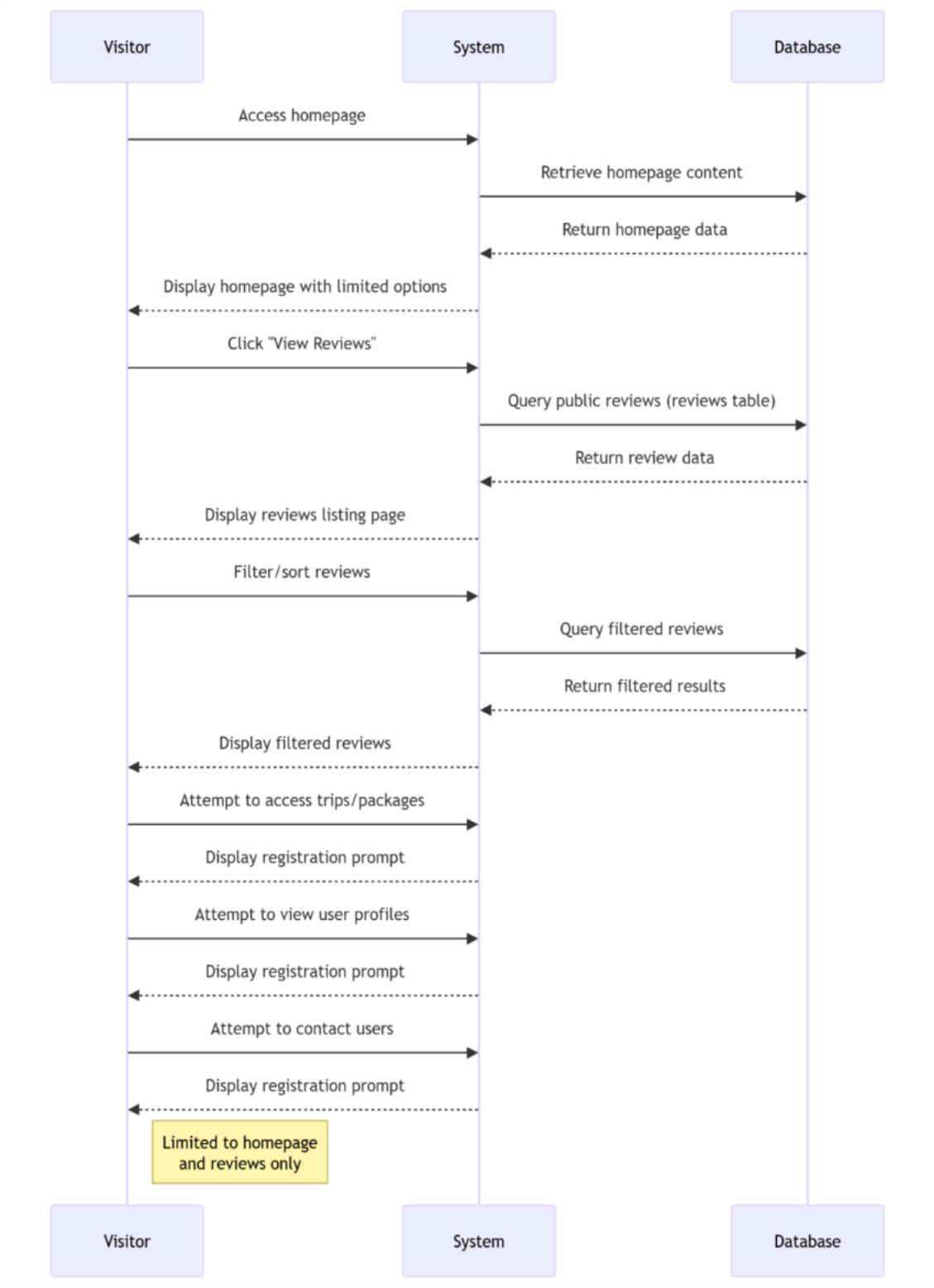

Login
Check
Invalid
Valid
User
Packages
Trip
Payment
Review
Group
Notification
Itinerary
Content Administration
View
Edit
Suspend
Add
Edit
Delete
View History
View Active Trips
View Upcoming Trips
View Confirm Payments
View Cancel Payments
View Pending Payments
View
Edit
Approve
Reject
View
Inquiry
View
Edit
Logout

CLASS DIAGRAM

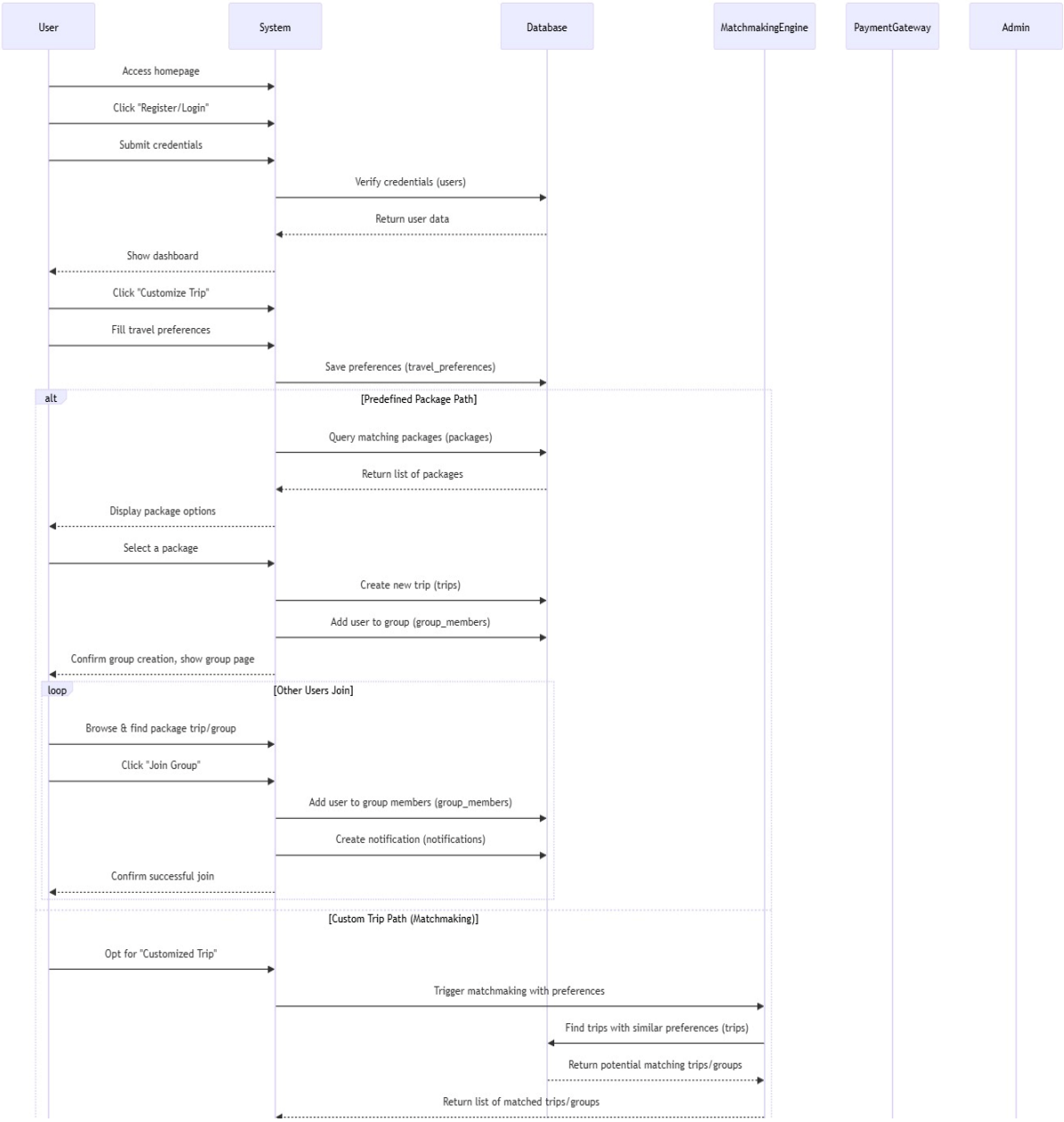


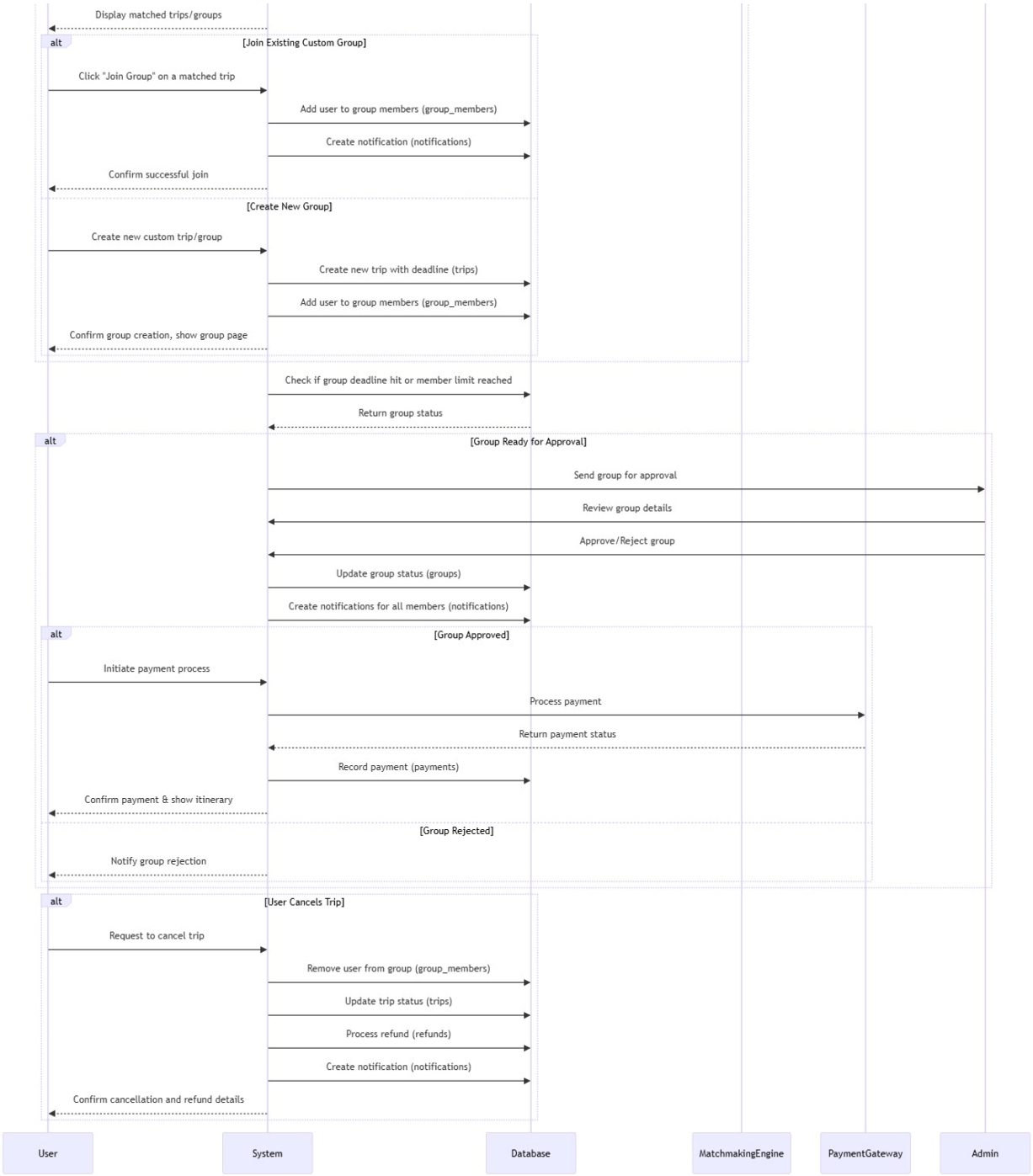
SEQUENCE DIAGRAM

1. For Visitor

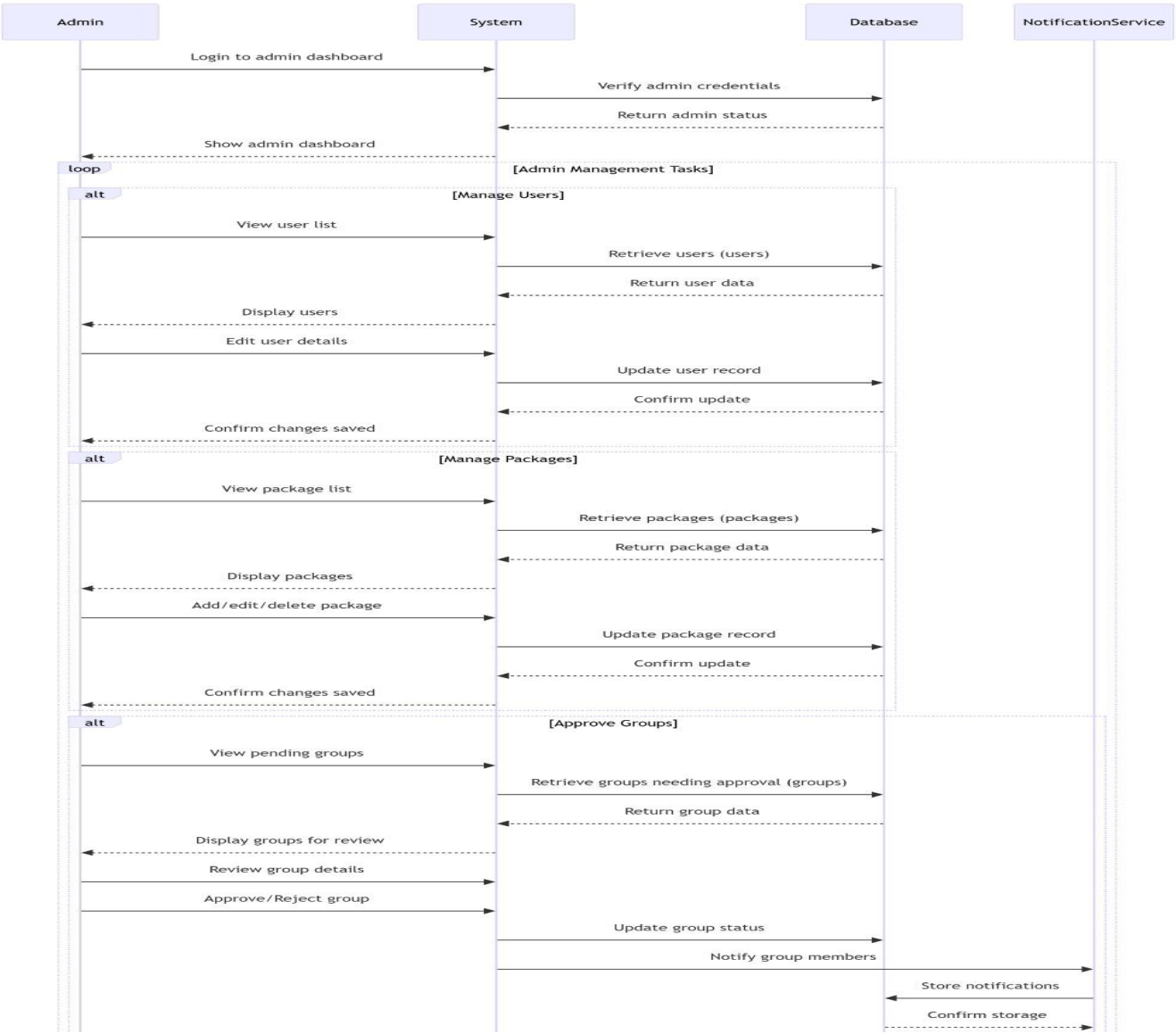


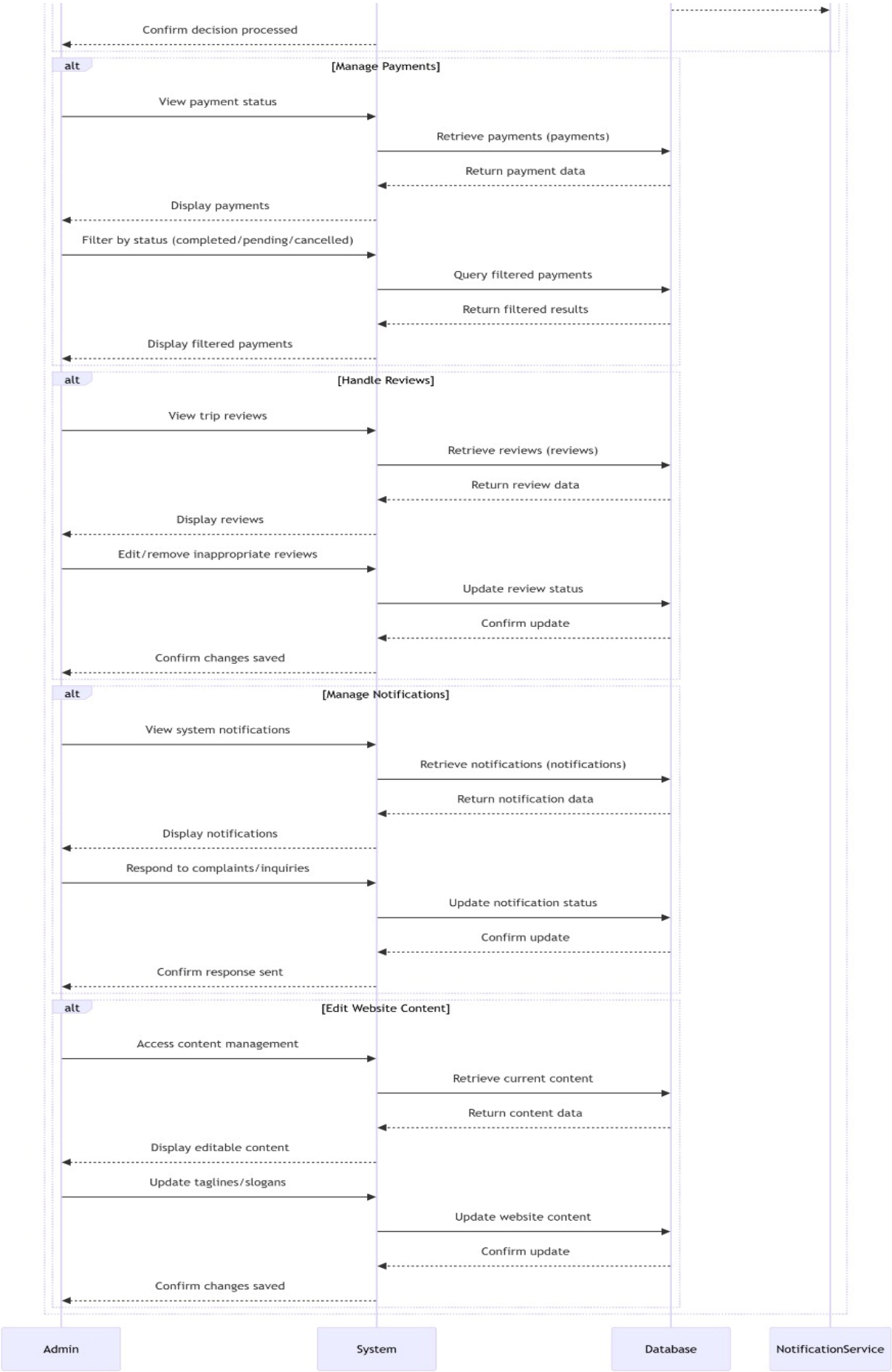
2. For User


User
System
Database
MatchmakingEngine
PaymentGateway
Admin
Access homepage
Click "Register/Login"
Submit credentials
Verify credentials (users)
Return user data
Show dashboard
Click "Customize Trip"
Fill travel preferences
Save preferences (travel_preferences)
alt
[Predefined Package Path]
Query matching packages (packages)
Return list of packages
Display package options
Select a package
Create new trip (trips)
Add user to group (group_members)
Confirm group creation, show group page
loop
[Other Users Join]
Browse & find package trip/group
Click "Join Group"
Add user to group members (group_members)
Create notification (notifications)
Confirm successful join
[Custom Trip Path (Matchmaking)]
Opt for "Customized Trip"
Trigger matchmaking with preferences
Find trips with similar preferences (trips)
Return potential matching trips/groups
Return list of matched trips/groups


Display matched trips/groups
alt
[Join Existing Custom Group]
Click "Join Group" on a matched trip
Add user to group members (group_members)
Create notification (notifications)
Confirm successful join
[Create New Group]
Create new custom trip/group
Create new trip with deadline (trips)
Add user to group members (group_members)
Confirm group creation, show group page
Check if group deadline hit or member limit reached
Return group status
alt
[Group Ready for Approval]
Send group for approval
Review group details
Approve/Reject group
Update group status (groups)
Create notifications for all members (notifications)
alt
[Group Approved]
Initiate payment process
Process payment
Return payment status
Record payment (payments)
Confirm payment & show itinerary
[Group Rejected]
Notify group rejection
alt
[User Cancels Trip]
Request to cancel trip
Remove user from group (group_members)
Update trip status (trips)
Process refund (refunds)
Create notification (notifications)
Confirm cancellation and refund details
User
System
Database
MatchmakingEngine
PaymentGateway
Admin

3.ADMIN


Admin
System
Database
NotificationService
Login to admin dashboard
Verify admin credentials
Return admin status
Show admin dashboard
loop
[Admin Management Tasks]
alt
[Manage Users]
View user list
Retrieve users (users)
Return user data
Display users
Edit user details
Update user record
Confirm update
Confirm changes saved
alt
[Manage Packages]
View package list
Retrieve packages (packages)
Return package data
Display packages
Add/edit/delete package
Update package record
Confirm update
Confirm changes saved
alt
[Approve Groups]
View pending groups
Retrieve groups needing approval (groups)
Return group data
Display groups for review
Review group details
Approve/Reject group
Update group status
Notify group members
Store notifications
Confirm storage


Confirm decision processed
alt
[Manage Payments]
View payment status
Retrieve payments (payments)
Return payment data
Display payments
Filter by status (completed/pending/cancelled)
Query filtered payments
Return filtered results
Display filtered payments
alt
[Handle Reviews]
View trip reviews
Retrieve reviews (reviews)
Return review data
Display reviews
Edit/remove inappropriate reviews
Update review status
Confirm update
Confirm changes saved
alt
[Manage Notifications]
View system notifications
Retrieve notifications (notifications)
Return notification data
Display notifications
Respond to complaints/inquiries
Update notification status
Confirm update
Confirm response sent
alt
[Edit Website Content]
Access content management
Retrieve current content
Return content data
Display editable content
Update taglines/slogans
Update website content
Confirm update
Confirm changes saved
Admin
System
Database
NotificationService

DATABASE DESIGN

1. USER TABLE

Field	Type	Description
user_id	Number (Auto Increment)	Primary key
name	String	User's full name
email	String (Unique)	User's email
phone number	String (Unique)	User's phone number
password	String	Hashed password
profile picture	String	Path to profile image
dob	Date	Date of birth
gender	Enum('Male','Female','Other')	Gender
bio	String	Biography
location	String	Location
stamin	Boolean	Admin status
last login	DateTime	Last login timestamp
is_active	Boolean	Active status
failed_attempts	Number	Failed login attempts

2. TRAVEL PREFERENCE

Field	Type	Description
preference_id	Number (Auto Increment)	Primary Key
user_id	Number (Ref: Users)	User Reference
destination_type	String	Preferred destination type
budget_range	String	Budget range
travel_type	Enum ('Solo','Group','Family')	Type of travel
interests	String	User's travel interests

3. PACKAGES

Field	Type	Description
package_id	Number (Auto Increment)	Primary key
title	String	Package title
description	String	Package description
price	Decimal	Package price
duration_days	Number	Duration of trip

4. TRIPS

Field	Type	Description
trip_id	Number(Auto Increment)	Primary key
user_id	Number(Ref:users)	User reference
destination	String	Destination location
start_date	Date	Trip start date
end_date	Date	Trip end date
status	Enum('Planned','Completed','Cancelled')	Trip status
notes	String	Additional notes
package_id	Number(Ref:packages)	Package references
created_at	DateTime	Creation timestamp
updated_at	DateTime	Last updated timestamp

5. GROUPS

Field	Type	Description
group_id	Number (Auto Increment)	Primary key
trip_id	Number (Ref:trips)	Trip reference
created_by	Number (Ref:users)	Creator user interface
description	String	Group description
no_of_members	Number	Number of members
created_at	DateTime	Creation timestamp

6. **GROUP MEMBER**

Field	Type	Description
group_id	Number(Ref:groups)	Group reference
user_id	Number (Ref:users)	User reference
joined_at	DateTime	Joined timestamp

7. **GROUP CHATS**

Field	Type	Description
message_id	Number(Auto Increment)	Primary key
group_id	Number (Ref:groups)	Group reference
sender_id	Number(Ref:users)	Sender reference
content	String	Chat message content
sent_at	DateTime	Message timestamp

8. **MATCHMAKING**

Field	Type	Description
match_id	Number (Auto Increment)	Primary key
trip_id	Number(Ref:trips)	Trip reference
user_id	Number(Ref:users)	User reference
travel_pref_id	Number(Ref:travel_preferences	Travel preference reference

9. PAYMENTS

Field	Type	Description
payment_id	Number (Auto Increment)	Primary key
user_id	Number (Ref:users)	User reference
trip_id	Number(Ref:trips)	Trip reference
amount	Decimal	Payment amount
method	String	Payment method
status	Enum('Pending,'Completed','Failed')	Payment status
paid_at	DateTime	Payment timestamp
reference_no	String	Payment reference number

10. REFUNDS

Field	Type	Description
refund_id	Number (Auto Increment)	Primary key
payment_id	Number (Ref:payments)	Payment reference
user_id	Number (Ref:users)	User reference
amount	Decimal	Refund amount
reason	String	Refund reason
requested_at	DateTime	Requested timestamp
received_at	DateTime	Received timestamp
processed_at	DateTime	Processed timestamp

11.REVIEWS

Field	Type	Description
review_id	Number (Auto Increment)	Primary key
user_id	Number (Ref:users)	User reference
trip_id	Number (Ref:trips)	Trip reference
rating	Number	Rating given
comment	String	Review comment
created_at	DateTime	Creation timestamp

12.TRAVEL HISTORY

Field	Type	Description
history_id	Number (Auto Increment)	Primary key
user_id	Number (Ref:users)	User reference
trip_id	Number (Ref:trips)	Trip reference
trip_date	DateTime	Date of trip
feedback	String	User feedback

13. NOTIFICATION

Field	Type	Description
notification_id	Number (Auto Increment)	Primary key
user_id	Number (Ref:users)	User reference
title	String	Notification title
message	String	Notification message
type	Enum('trip','review','message','admin_alert')	Notification type
is_read	Boolean	Read status
created_at	DateTime	Created timestamp

4. USER INTERFACE DESIGN

DESCRIPTION OF INTERFACE

- User Interface Design

The TripConnect UI is designed to be simple, responsive, and user-friendly, allowing users to easily customize trips, join groups, and manage bookings.

- Description of the User Interface

Homepage: Tagline, navigation menu (Home, Packages, Reviews, Contact), Login/Sign Up, and "Customize Trip" button.

- Preferences Form: Collects user travel details (destination, budget, interests).
- Group & Notifications: Shows group status, approvals, alerts, and chat.
- Admin Dashboard: For reviewing and approving groups, managing payments, and monitoring activities.
- Screen Images

Key screens include the Homepage, Preferences Form, Group Creation Page, and Notifications Panel.

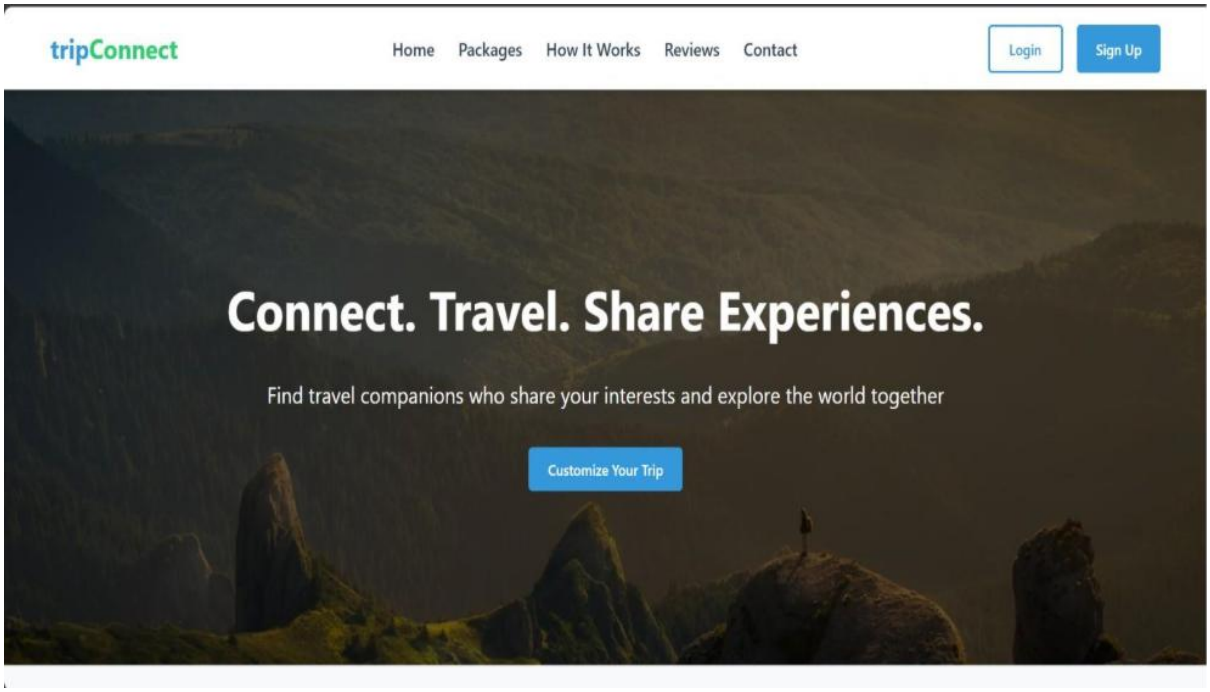
- Objects and Actions

Buttons (Login, Sign Up, Customize Trip): Navigate or submit data.

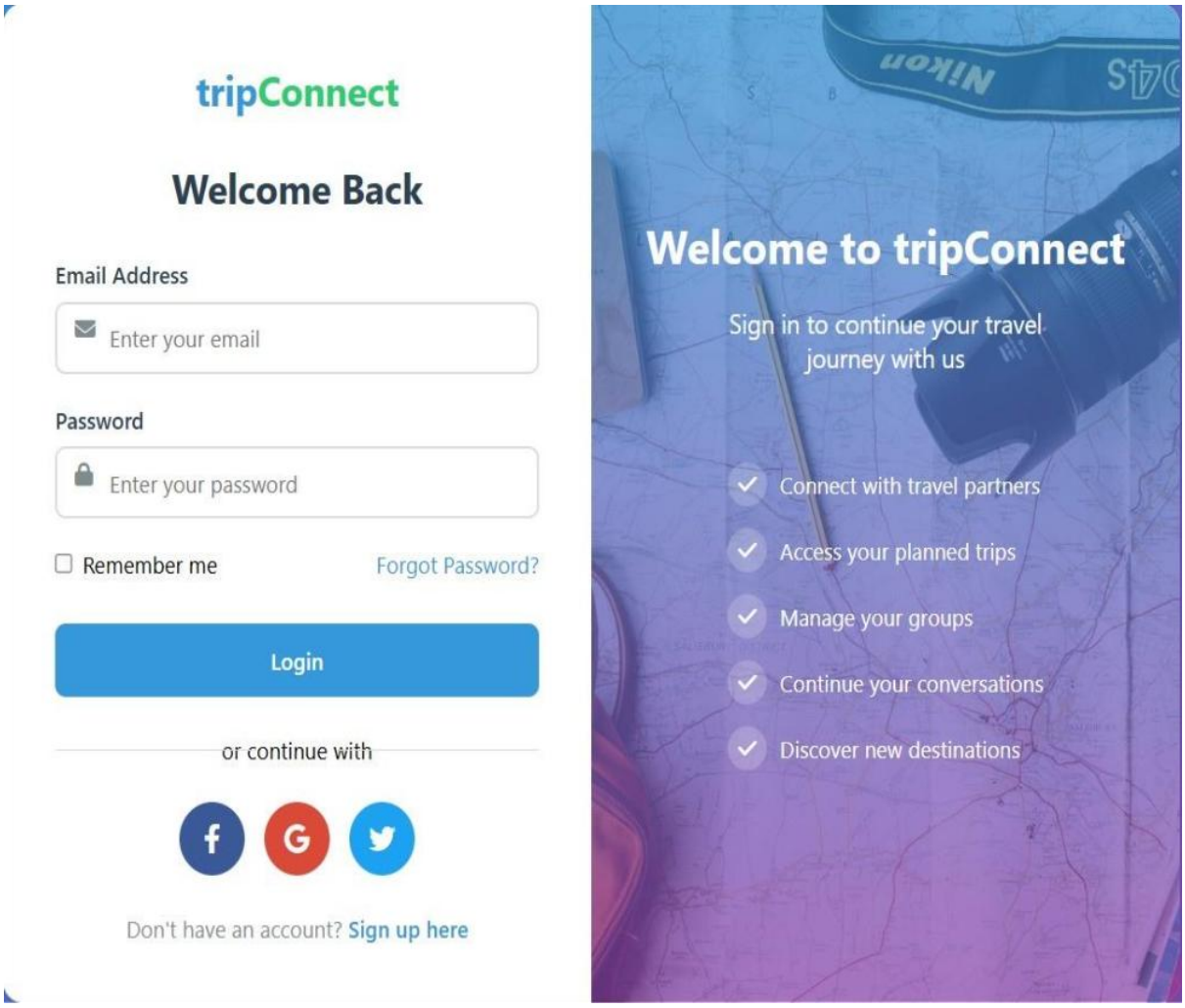
- Text Fields: Capture user details like name, email, and preferences.
- Dropdowns: Select travel type (Solo/Group/Family).
- Notification Icons: Show alerts for approvals, payments, and messages.
- Chat Box: Allows group members to exchange messages.

4.1 SCREEN IMAGES

1. HOMEPAGE

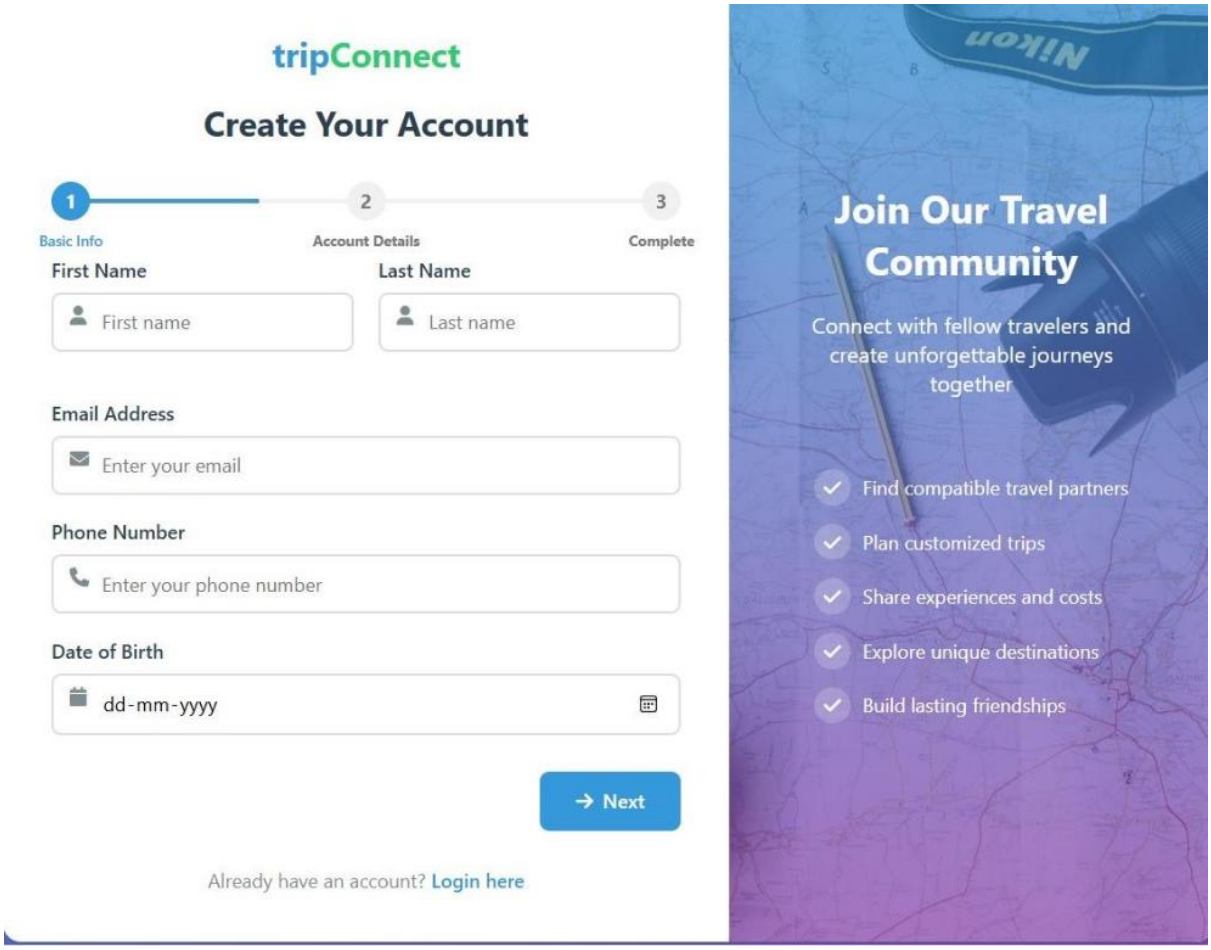


3. LOGIN PAGE

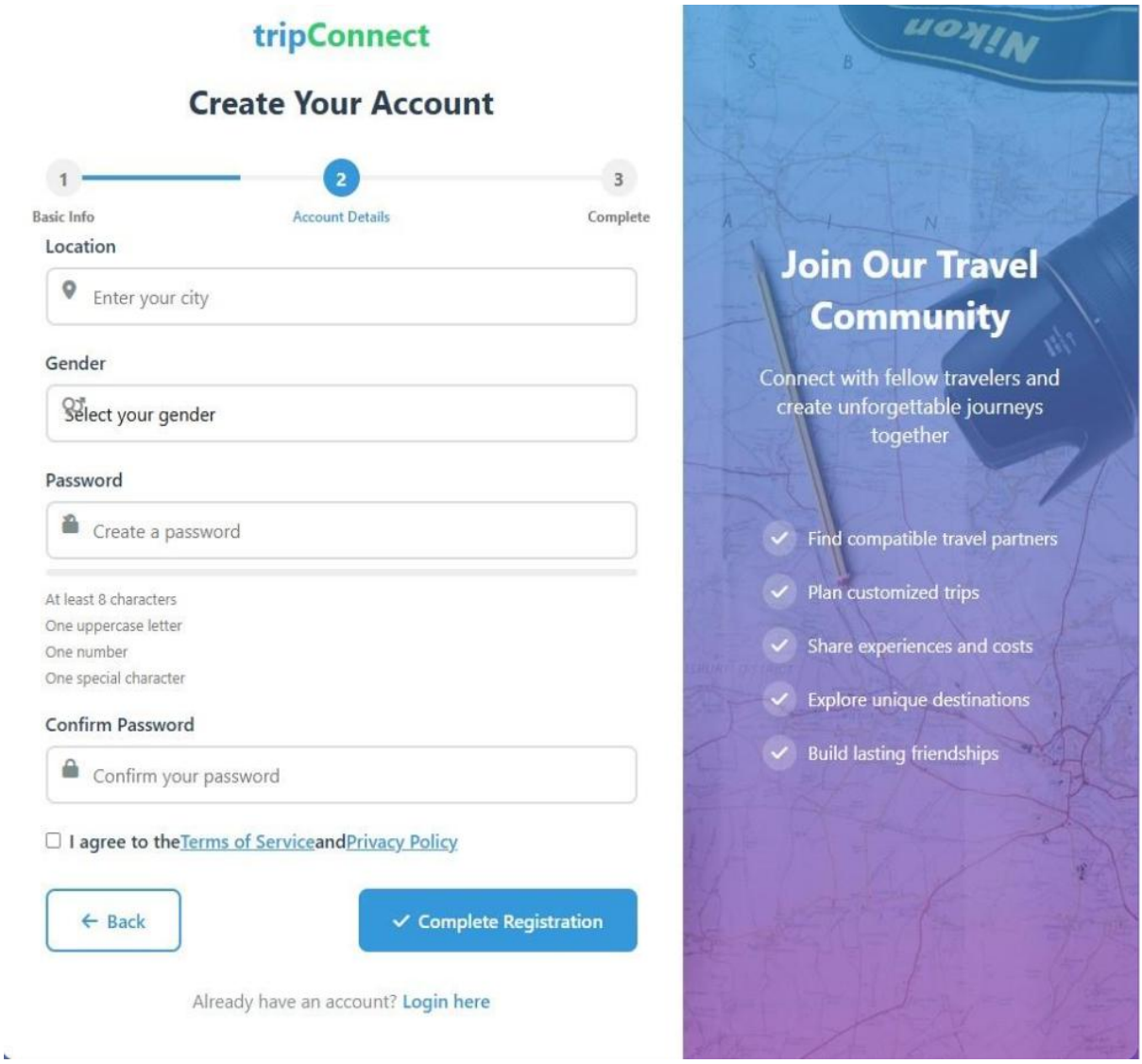

tripConnect
Welcome Back
Email Address
Enter your email
Password
Enter your password
Remember me
Forgot Password?
Login
or continue with
Don't have an account? Sign up here
Welcome to tripConnect
Sign in to continue your travel journey with us
Connect with travel partners
Access your planned trips
Manage your groups
Continue your conversations
Discover new destinations

3.REGISTER PAGE

3.1



3.2


tripConnect
Create Your Account
1
2
3
Basic Info
Account Details
Complete
Location
Enter your city
Gender
Select your gender
Password
Create a password
At least 8 characters
One uppercase letter
One number
One special character
Confirm Password
Confirm your password
I agree to theTerms of ServiceandPrivacy Policy
Back
Complete Registration
Already have an account? Login here
Join Our Travel Community
Connect with fellow travelers and create unforgettable journeys together
Find compatible travel partners
Plan customized trips
Share experiences and costs
Explore unique destinations
Build lasting friendships

- **TYPES OF TESTS**

Unit Tests

Unit tests are most commonly done by developers on their own machines or on a common server that is very volatile. It is not necessary that the unit test machines be the same platform and operating system as the target deployment environment, but the movement from the unit test environment to other testing environments should not require material code changes by developers. A plan for one machine per developer plus one small server should be included in the overall system architecture.

System Tests

The system test environment allows multiple modules to be connected together and executed as in a typical use-case scenario. The choice as to whether this is done on a separate machine from unit testing is up to the implementation and test team. If the target deployment environment is different from the unit test environment, the system test environment should contain a machine that matches the target environment. Although the system test machine need not match the size of the deployment box, it should have the same platform and operating system. A good rule of thumb is to prepare to add one more box for system tests of a smaller size, but the same operating system as the target environment. Again, this will be a relatively volatile environment, so it should not be viewed as a place to do industrial strength testing by a large team.

Integration or Regression Tests

To perform integration and regression tests, it is advisable to have a separate environment that is similar to the target environment. Generally, one server will be enough at this point. However, the contents of this server should be strictly controlled. Either the test coordinator or his or her designate should make all software changes to this environment. Stability and auditability are essential to ensuring the accuracy of test results. Plan for at least one more servers at this stage in testing.

Stress Tests

Stress tests should be done in an environment identical to the target hosting environment. In the first development cycle, this can be done in the production site, before the cutover to production. For subsequent development cycles, a separate environment will have to be maintained for stress testing. Plan to replicate the deployment environment as part of the test bed for at least the second development cycle of the site. It has also been observed that the most common problem after performance in a high stress environment is database deadlock due to improper programming. Deadlocks are typically difficult to detect and fix and may not show up until the site is highly stressed in production. So it is important that these conditions be stringently tested during the stress test phase.

Acceptance Tests and Staging

Acceptance tests are generally performed in the same environment as the stress tests, so additional hardware is not needed to support this phase of testing. Again, during the initial development cycle, the production environment can be used to perform both acceptance testing and staging, but a new environment should be created for subsequent development cycles.

REFERENCES

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- Sommerville, Ian. Software Engineering, 10th Edition, Pearson, 2016.

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